

Linear system: $L[au + bv] = aL[u] + bL[v]$, $a, b \in \mathbb{R}$

LTI: $y(t) = L[x(t)] \implies y(t - t_0) = Lx(t - t_0)$

Delta function:

1. $\int \delta(t) dt = 1$
2. $\delta(t) = 0$ for $t \neq 0$
3. $\int f(t)\delta(t - t_0) dt = f(t_0)$ (sifting)

Impulse response: the output of an LTI system when the input is $\delta(t)$

Transfer function: $H(f) = \mathcal{F}(h(t))$ where $h(t)$ is the impulse response of the system.

$$\delta(t - \tau) \quad \text{---} \boxed{h(t)} \text{---} \quad h(t)$$

where the box represents a **convolution**:

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

(in fact, in the frequency domain, this operation is defined by $Y(f) = X(f)H(f)$)

Fourier Transform: $\mathcal{F}[x(t)] = \int x(t)e^{-j2\pi ft} dt = X(f)$

Inverse Fourier: $\mathcal{F}^{-1}[X(f)] = \int X(f)e^{j2\pi ft} df = x(t)$

Standard Fourier pairs:

1. $\mathcal{F}[\delta(t)] = 1$
2. $\mathcal{F}[e^{-j2\pi f_0 t}] = \delta(f + f_0)$
3. $\mathcal{F}[\delta(t - t_0)] = e^{-j2\pi f t_0}$
4. $X(f - f_0) = x(t)e^{j2\pi f_0 t}$
5. $\mathcal{F}[\cos 2\pi f_0 t] = \frac{1}{2}[\delta(f + f_0) + \delta(f - f_0)]$
6. $\mathcal{F}[\sin 2\pi f_0 t] = \frac{1}{2j}[\delta(f + f_0) - \delta(f - f_0)]$
7. $\mathcal{F}[\mathbb{1}_{x \in [-T/2, T/2]}] = \text{sinc}(2\pi T f)$

Fourier Properties:

1. $\mathcal{F}[x(t - t_0)] = e^{-j2\pi f t_0} X(f)$
2. $\mathcal{F}[x(at)] = \frac{1}{|a|} X(f/a)$
3. $\mathcal{F}[x'(t)] = X(f) \cdot j2\pi f$
4. $\mathcal{F}[\int_{-\infty}^t x(u) du] = \frac{X(f)}{j2\pi f} + \frac{X(0)}{2} \delta(f)$
5. $\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$ (Parseval's)
6. $\mathcal{F}[x(-t)] = X(-f)$
7. $\mathcal{F}[X(f)] = x(-t)$

Useful tricks:

1. $|X(f)|^2 = X(f)X^*(f) = X(f)X(-f)$
2. $\cos(2\pi ft) = (e^{j2\pi ft} + e^{-j2\pi ft})/2$

$$3. \sin(2\pi ft) = (e^{j2\pi ft} - e^{-j2\pi ft})/2j$$

4. Fourier of odd function is purely imaginary, Fourier of even function is purely real.

5. δ is an even function: $\delta(-t) = \delta(t)$

$$6. u(t) = \mathbb{1}_{t \geq 0}(t)$$

$$7. \sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$8. \cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$9. \sin^2 x = (1 - \cos 2x)/2$$

$$10. \cos^2 x = (1 + \cos 2x)/2$$

$$11. \cos \theta(t) = \sum_{m=0}^{\infty} \frac{(\theta(t))^{2m}}{(2m)!}$$

$$12. \sin \theta(t) = \sum_{m=0}^{\infty} (-1)^m \frac{(\theta(t))^{2m+1}}{(2m+1)!}$$

$$13. \mathcal{F}[x(t)y(t)] = X(f) * Y(f)$$

Baseband: unmodulated signal (i.e. the message signal).

Passband: modulated signal (i.e. the transmitted signal).

AM Coherent modulation: $r(t) = m(t) \cos 2\pi f_c t$, demodulate with $\cos 2\pi f_c t$ and LPF to recover $m(t)/2$.

AM Large carrier: $r(t) = (m(t) + A) \cos 2\pi f_c t$ ($A \gg m(t)$), demodulate with an envelope detector, LPF, and HPF to recover $m(t)$.

AM Single sideband: $r(t) = m(t) \cos 2\pi f_c t$ with an LPF applied to $m(t)$ before modulation (takes advantage of symmetry from $M(-f) = M^*(f)$ to use half the bandwidth), demodulate with $\cos 2\pi f_c t$ and LPF to recover $m(t)/2$.

Frequency modulation: $r(t) = \sin(2\pi f_c t + \beta \int_{-\infty}^t m(\tau) d\tau)$, demodulate by differentiating, envelope detecting, and HPF to recover $\beta m(t)$.

Phase modulation: $r(t) = \sin(2\pi f_c t + \beta m(t))$ (filter and integrate)

Carson's Rule: $\Delta = (\kappa + 1)B$ where Δ is the single-sided FM bandwidth, κ is the modulation index, and B is the bandwidth of the message signal.

In particular,

$$\begin{aligned} r(t) &= \cos(2\pi f_c t + \beta m(t)) \\ &= \cos(2\pi f_c t) \cos(\beta m(t)) - \sin(2\pi f_c t) \sin(\beta m(t)) \\ &= \cos(2\pi f_c t) \sum_{m=0}^{\infty} \frac{(-1)^m (\beta m(t))^{2m}}{(2m)!} \\ &\quad - \sin(2\pi f_c t) \sum_{m=0}^{\infty} \frac{(-1)^m (\beta m(t))^{2m+1}}{(2m+1)!} \end{aligned}$$

where the Taylor series terms are small for $m = e\beta > e\beta(2\pi m)^{-1/(2m)}$ (via Stirling's approximation). Hence taking the first $N = (e/2)\beta$ terms in the Taylor expansion gives a good approximation of the bandwidth of the FM signal ($\approx m^{2N-1}(t) \sin 2\pi f_c t \implies \Delta \approx 2B(2N - 1) = 2B(e\beta - 1)$).